APEX NCAA FOOTBALL - DAILY GRADER

TARGET SLATE 2026-10-21

Layer 1 (sections A-F): yesterday, season record, probability quality against FanDuel, calibration, ratings, integrity. Layer 2 (appendices 1-5): technical evidence, hashes and diagnostics.

FIELD	PHYSICAL VALUE
TARGET SLATE	2026-10-21
GENERATED ET	2026-10-22T07:01:23.544234-04:00
ATS MODEL	D2 / d50e21788ae9f7107b8641e1e8e89daad26b3e828b85994137d2fe576346fd19
TOTALS MODEL	TOTALS-V5-RECONCILED-POOL / 3a38517a5262c5e818e8f3222ff0811d748e4c6e03cccc7950ffecc626149980
CANONICAL ISSUANCE	3840287e23fc5e6008b90b4d6b390ffe391c9b85b60ce8f23759df3897315107
GRADING COMPLETION RECEIPT	7a5eca5899a06278789c8bd281a272d790fe38696b92e0590f42a317f96e4a17
REPORT SCHEMA	apex.ncaaf.daily_grader_evidence_book.v5.two_layer

POPULATIONS USED IN THIS REPORT

POPULATION	DEFINITION	SLATES	GAMES	ATS	TOTALS	POSITIONS
TARGET SLATE	all physical positions issued for 2026-10-21	1	1	1	1	2
of which EXCLUDED POST-KICKOFF	issued after kickoff; visible, never scored	—	—	—	—	0
of which PROSPECTIVE ELIGIBLE	enter records and proper scores	—	—	1	1	2
PHYSICAL POSITION-EVIDENCE CORPUS	every immutable issued position through 2026-10-21	30	510	510	510	1020
minus EXCLUDED POST-KICKOFF	issued after kickoff; visible, never scored	2	16	16	16	32
= RECORD ELIGIBLE GRADED CORPUS	the only population in records, calibration and scores	30	494	494	494	988
settled (W+L+P+V)	—	—	—	494	494	988
pending	official result not final	—	—	0	0	0

RECONCILIATION · physical 1020 – excluded post-kickoff 32 = record eligible 988 (ATS 510 – 16 = 494; TOTALS 510 – 16 = 494). Excluded by slate: 2026-09-25 (2), 2026-09-26 (30).

HOW TO READ

Records use W, L, P (push), V (void) and pending. Win rate is W/(W+L).

Every grade settles the exact immutable issued proposition. No engine is rerun and no later FanDuel line replaces the issued line.

Ratings (WEAK / MODERATE / STRONG / ELITE) are labels on the selected win probability, assigned at issuance by the rating definition sealed with each position (tier_policy_sha256) and retained exactly as issued, except a separately recorded and verified pre-kickoff public revision. Original ratings and revision evidence remain in the position export; no grader run computes new ratings. The definition changed during the season, so a label has one meaning only within one definition and engine. Target-slate positions were rated under SLATE-QUARTILES-V3. A rating is relative confidence within its definition and engine, not a win probability; the model probability is shown separately (APEX P). Ratings are evaluated, not assumed, in section E, grouped by definition.

RATING DEFINITION FOUR-TIER-V1 · APEX_NCAAF_CANONICAL_FOUR_TIER_V1 · sha256 65fbd6df6e5528e7… · issued 2026-08-29 … 2026-09-03 · 38 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5300 / 0.5600 / 0.6000 · TOTALS 0.5300 / 0.5600 / 0.6000 (same cut points for both engines).

RATING DEFINITION SHARED-51.75 · APEX_MAX_CALIBRATED_WIN_PROBABILITY_RATING_CONTRACT_V3_PRESENTATION_MIRROR · sha256 c7d506cdb615c0db… · issued 2026-09-04 … 2026-09-19 · 476 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5175 / 0.5375 / 0.5575 · TOTALS 0.5175 / 0.5375 / 0.5575 (same cut points for both engines).

RATING DEFINITION SHARED-52/54/58 · APEX_MAX_CALIBRATED_WIN_PROBABILITY_RATING_CONTRACT_V3_PRESENTATION_MIRROR · sha256 a69977c71f3df62b… · issued 2026-09-24 … 2026-10-09 · 282 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5200 / 0.5400 / 0.5800 · TOTALS 0.5200 / 0.5400 / 0.5800 (same cut points for both engines).

RATING DEFINITION ENGINE-V2 · APEX_NCAAF_ENGINE_SPECIFIC_RATING_BANDS_V2_20261009 · sha256 a7acf120b9b1c889… · issued 2026-10-10 … 2026-10-16 · 112 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS 0.5120 / 0.5318 / 0.5460 · TOTALS 0.5044 / 0.5111 / 0.5209 (engine-specific cut points at the 40th / 78th / 95th percentiles of a 100-position reference per engine (Oct 3 and Oct 10 rescore)).

RATING DEFINITION SLATE-QUARTILES-V3 · APEX_NCAAF_SLATE_RELATIVE_QUARTILES_V3 · sha256 6de90770392f7efd… · issued 2026-10-17 … 2026-10-21 · 112 positions · MODERATE / STRONG / ELITE lower cut points on the selected win probability: ATS slate-specific (retained per position) · TOTALS slate-specific (retained per position) (strict-lower percentile rank within each engine's scoreable slate; rank boundaries 0.25 / 0.50 / 0.75; probability cut points vary by slate and are retained per position; ties share a tier).

NCAA full-game spread and total settlement includes overtime unless the applicable issued-market rules state otherwise. A whole-number line landing on the result is a push.

KNOWN EVIDENCE LIMITATION · FANDUEL_SETTLEMENT_RULE_SOURCE=UNRECORDED_JURISDICTION_DETAIL: the jurisdiction-specific FanDuel settlement rule text in force for each issued market is not recorded in the evidence; every grade settles the issued line and price against the official final score.

A. YESTERDAY — TARGET-SLATE POSITION LEDGER (2026-10-21)

2 physical positions on 1 games; 0 excluded post-kickoff. FD P = same-side FanDuel no-vig probability from the issuance-time quote (two-way multiplicative normalization); Δ = APEX – FD.

G	MATCHUP	KICK ET	MKT	SELECTION	LINE	PRICE	APEX P	FD P	Δ	RATING	FINAL	RES
1	New Mexico State at Sam Houston	Wed 20:00	ATS	New Mexico State	-2.5	-120	52.3%	51.9%	+0.4	WEAK	30-24 (A-H)	W
1	New Mexico State at Sam Houston	Wed 20:00	TOTALS	OVER	46.5	-120	51.0%	51.9%	-1.0	WEAK	30-24 (A-H)	W

TARGET-SLATE RECORD · ATS 1W-0L-0P-0V (100.0%) · TOTALS 1W-0L-0P-0V (100.0%) · COMBINED 2W-0L-0P-0V (100.0%) · pending 0 · excluded post-kickoff 0. All positions derive from immutable issuance 3840287e23fc5e60…; no closing-line substitution.

B. SEASON — RECORD ELIGIBLE GRADED CORPUS

30 physical slates 2026-08-29 … 2026-10-21; 988 record eligible positions. Win rate is an operational outcome, not the forecast evaluation (see C and D).

ENGINE	N	W	L	P	V	PENDING	SETTLED	WIN %
ATS	494	260	234	0	0	0	494	52.6%
TOTALS	494	254	240	0	0	0	494	51.4%
COMBINED	988	514	474	0	0	0	988	52.0%
COMBINED · FOUR-TIER-V1 · WEAK	23	16	7	0	0	0	23	69.6%
COMBINED · FOUR-TIER-V1 · MODERATE	5	1	4	0	0	0	5	20.0%
COMBINED · FOUR-TIER-V1 · STRONG	3	1	2	0	0	0	3	33.3%
COMBINED · FOUR-TIER-V1 · ELITE	7	6	1	0	0	0	7	85.7%
COMBINED · SHARED-51.75 · WEAK	195	108	87	0	0	0	195	55.4%
COMBINED · SHARED-51.75 · MODERATE	190	95	95	0	0	0	190	50.0%
COMBINED · SHARED-51.75 · STRONG	71	37	34	0	0	0	71	52.1%
COMBINED · SHARED-51.75 · ELITE	20	14	6	0	0	0	20	70.0%
COMBINED · SHARED-52/54/58 · WEAK	125	65	60	0	0	0	125	52.0%
COMBINED · SHARED-52/54/58 · MODERATE	85	33	52	0	0	0	85	38.8%
COMBINED · SHARED-52/54/58 · STRONG	40	22	18	0	0	0	40	55.0%
COMBINED · SHARED-52/54/58 · ELITE	0	0	0	0	0	0	0	N/A
COMBINED · ENGINE-V2 · WEAK	53	26	27	0	0	0	53	49.1%
COMBINED · ENGINE-V2 · MODERATE	43	20	23	0	0	0	43	46.5%
COMBINED · ENGINE-V2 · STRONG	15	8	7	0	0	0	15	53.3%
COMBINED · ENGINE-V2 · ELITE	1	1	0	0	0	0	1	100.0%
COMBINED · SLATE-QUARTILES-V3 · WEAK	30	15	15	0	0	0	30	50.0%
COMBINED · SLATE-QUARTILES-V3 · MODERATE	28	18	10	0	0	0	28	64.3%
COMBINED · SLATE-QUARTILES-V3 · STRONG	28	14	14	0	0	0	28	50.0%
COMBINED · SLATE-QUARTILES-V3 · ELITE	26	14	12	0	0	0	26	53.8%

SLATE	ATS N	ATS W-L-P	ATS WIN %	TOT N	TOT W-L-P	TOT WIN %	EXCL
2026-08-29	8	5-3-0	62.5%	8	6-2-0	75.0%	0
2026-09-03	11	9-2-0	81.8%	11	4-7-0	36.4%	0
2026-09-04	7	4-3-0	57.1%	7	4-3-0	57.1%	0
2026-09-05	68	39-29-0	57.4%	68	33-35-0	48.5%	0
2026-09-06	3	0-3-0	0.0%	3	2-1-0	66.7%	0
2026-09-07	1	1-0-0	100.0%	1	1-0-0	100.0%	0
2026-09-10	1	1-0-0	100.0%	1	0-1-0	0.0%	0
2026-09-11	5	3-2-0	60.0%	5	2-3-0	40.0%	0
2026-09-12	80	43-37-0	53.8%	80	44-36-0	55.0%	0
2026-09-17	1	1-0-0	100.0%	1	1-0-0	100.0%	0
2026-09-18	3	1-2-0	33.3%	3	1-2-0	33.3%	0
2026-09-19	69	33-36-0	47.8%	69	40-29-0	58.0%	0
2026-09-24	1	0-1-0	0.0%	1	1-0-0	100.0%	0
2026-09-25	4	2-2-0	50.0%	4	2-2-0	50.0%	2
2026-09-26	49	22-27-0	44.9%	49	25-24-0	51.0%	30
2026-10-01	2	1-1-0	50.0%	2	1-1-0	50.0%	0
2026-10-02	3	2-1-0	66.7%	3	1-2-0	33.3%	0
2026-10-03	54	29-25-0	53.7%	54	22-32-0	40.7%	0
2026-10-06	1	1-0-0	100.0%	1	0-1-0	0.0%	0
2026-10-07	2	1-1-0	50.0%	2	2-0-0	100.0%	0
2026-10-08	4	1-3-0	25.0%	4	3-1-0	75.0%	0
2026-10-09	5	3-2-0	60.0%	5	1-4-0	20.0%	0
2026-10-10	46	19-27-0	41.3%	46	24-22-0	52.2%	0
2026-10-13	2	2-0-0	100.0%	2	2-0-0	100.0%	0
2026-10-14	2	1-1-0	50.0%	2	1-1-0	50.0%	0
2026-10-15	3	2-1-0	66.7%	3	2-1-0	66.7%	0

SLATE	ATS N	ATS W-L-P	ATS WIN %	TOT N	TOT W-L-P	TOT WIN %	EXCL
2026-10-16	3	0-3-0	0.0%	3	2-1-0	66.7%	0
2026-10-17	52	30-22-0	57.7%	52	24-28-0	46.2%	0
2026-10-20	3	3-0-0	100.0%	3	2-1-0	66.7%	0
2026-10-21	1	1-0-0	100.0%	1	1-0-0	100.0%	0

RETURN IN UNITS (1 unit per pick at the issued price; push = 0) [R2][R3][R9] · ATS season 260-234 (52.6%) · -2.49 u · return -0.5%; yesterday 1-0 (100.0%) · +0.83 u · return 83.3% · TOTALS season 254-240 (51.4%) · -10.28 u · return -2.1%; yesterday 1-0 (100.0%) · +0.83 u · return 83.3% · COMBINED season 514-474 (52.0%) · -12.77 u · return -1.3%; yesterday 2-0 (100.0%) · +1.67 u · return 83.3%.

BREAK-EVEN at -110 is 52.38% [R1] · ATS season 52.6% · TOTALS season 51.4%.

BEST BETS — SPREAD PICKS IN THE TOP EDGE GROUP [R4][R5][R10]

SPREAD BEST BETS	EDGE AT LEAST	SEASON W-L	SEASON WIN %	SEASON UNITS	SEASON RETURN	YESTERDAY W-L	YESTERDAY UNITS
TOP 10% BY EDGE	+0.0292	30-18	62.5%	+9.13	19.0%	0-0	+0.00
TOP 25% BY EDGE	+0.0215	69-52	57.0%	+10.30	8.5%	0-0	+0.00

Edge = APEX P − same-side FanDuel no-vig P from the quote at issuance. The top-10% / top-25% marks are taken over all graded spread picks this season. Grader view only: picks, ratings, the pick card and the website are unchanged. Small groups; see the sample-size line.

CLOSING LINE (FanDuel about 5 minutes before kickoff, Odds API history) [R8][R9] standard practice · ATS beat the close 216 of 475 (45.5%), same 46, worse 213, average +0.42 pts (yesterday 1 of 1) · TOTALS beat the close 219 of 475 (46.1%), same 44, worse 212, average +0.07 pts (yesterday 0 of 1). Grading still uses the issued line.

SAMPLE SIZE [R5][R9] · ATS 494 graded bets of the 500-1,200 research says are needed to judge skill · TOTALS 494 graded bets of the 500-1,200 research says are needed to judge skill.

SHADOW RULES — SIDE BY SIDE ONLY, LIVE PICKS UNCHANGED [R4][R5][R7][R10]

BEST BET (spread, edge at or above the top 15% of EARLIER 2026 spread edges, needs 50 earlier; today's mark +0.0260) · season 57-40 (58.8%) · +11.55 u · return 11.9% · yesterday 0-0 (—) · +0.00 u · return —

NO PLAY (edge 0 or less) · ATS 36-33 (52.2%) · -2.19 u · return -3.2% · TOTALS 34-24 (58.6%) · +5.54 u · return 9.6% · yesterday 1-0

TOTALS BRIER yesterday: APEX 0.2405 vs FanDuel 0.2311 (N=1, lower is better)

TOTALS BRIER season: APEX 0.2512 vs FanDuel 0.2489 (N=486, lower is better)

C. PROBABILITY QUALITY AGAINST FANDUEL — PAIRED, IDENTICAL OBSERVATIONS

Each row pairs the APEX issued probability with the same-side FanDuel no-vig probability from the quote captured at issuance, on the same game, market and settled W/L outcome. Lower is better. Δ = APEX − FanDuel (negative favours APEX).

ENGINE	PAIRED N	APEX BRIER	FD BRIER	Δ BRIER	APEX LOGLOSS	FD LOGLOSS	Δ LOGLOSS	APEX < FD (BRIER)
ATS	486	0.249890	0.250597	-0.000707	0.692926	0.694343	-0.001417	252 of 486
TOTALS	486	0.251178	0.248937	+0.002240	0.695503	0.691022	+0.004481	238 of 486

BRIER VS FANDUEL, IN ONE LINE (lower is better) [R6][R7] · ATS APEX better (APEX 0.2499 vs FanDuel 0.2506, N=486) · TOTALS FanDuel better (APEX 0.2512 vs FanDuel 0.2489, N=486).

Record eligible W/L rows without a FanDuel comparator in their immutable grade (not in the paired table, listed by slate and sealed engine): ATS 8 [2026-08-29 (8, GLOBAL_EMPIRICAL_V1)]; TOTALS 8 [2026-08-29 (8, RIDGE_GAUSSIAN_A100_V1)]. FanDuel no-vig method as registered in each grade object: TWO_WAY_MULTIPLICATIVE_NORMALIZATION.

TOTALS — RETAINED FULL-DISTRIBUTION SCORES (APEX ONLY)

SCORE	N	VALUE	CONVENTION
PMF LOGSCORE (total PMF)	491	4.081490	retained integer-support PMF, read from the immutable grade
RPS	491	0.034207	retained integer-support PMF, read from the immutable grade
ISSUED PUSH PROBABILITY (mean)	494	0.000000	explicit in each sealed distribution

No FanDuel full-distribution comparator exists: a single total line and two prices do not determine a distribution over totals, so none is constructed. The paired FanDuel comparison above is on the binary selected-side probability only.

D. CALIBRATION (RECORD ELIGIBLE W/L ROWS)

Calibration (is p the realized frequency?) and discrimination (does p separate outcomes?) are reported separately. Brier = miscalibration − discrimination + uncertainty (Murphy decomposition on the PAV fit).

ENGINE	N	MEAN ISSUED P	EMPIRICAL FREQ	CAL GAP	BRIER	LOGLOSS	MISCAL (MCB)	DISCRIM (DSC)	UNCERT (UNC)	AUC
ATS	494	0.5203	0.5263	+0.0061	0.248963	0.690673	0.001655	0.003034	0.249307	0.4797
TOTALS	494	0.5240	0.5142	-0.0098	0.250916	0.695217	0.001981	0.003428	0.249799	0.4667

ATS DIAGNOSTIC BIN (issued p)	N	W	L	MEAN P	EMPIRICAL	GAP
[0.50, 0.51)	156	86	70	0.5051	0.5513	+0.0461
[0.51, 0.52)	133	72	61	0.5152	0.5414	+0.0261
[0.52, 0.54)	172	80	92	0.5283	0.4651	-0.0632
[0.54, 0.58)	31	20	11	0.5497	0.6452	+0.0955
[0.58, 1.00)	2	2	0	0.8848	1.0000	+0.1152
TOTALS DIAGNOSTIC BIN (issued p)	**N**	**W**	**L**	**MEAN P**	**EMPIRICAL**	**GAP**
[0.50, 0.51)	158	87	71	0.5044	0.5506	+0.0463

TOTALS DIAGNOSTIC BIN (issued p)	N	W	L	MEAN P	EMPIRICAL	GAP
[0.51, 0.52)	122	64	58	0.5140	0.5246	+0.0106
[0.52, 0.54)	119	53	66	0.5291	0.4454	-0.0837
[0.54, 0.58)	82	41	41	0.5513	0.5000	-0.0513
[0.58, 1.00)	13	9	4	0.6373	0.6923	+0.0550

DIAGNOSTIC BINS ONLY · fixed reporting edges 0.50 / 0.51 / 0.52 / 0.54 / 0.58 on the issued probability, the same every day so bins compare across days. They are not rating cut points: each position's rating comes from its own sealed definition (HOW TO READ and section E). Small bins are shown with their N; nothing is pooled across engines.

CALIBRATION AVERAGE MISS (bin-weighted average gap between predicted and actual win rate; research target under 5 pts) [R6][R7] · ATS 5.0 pts (N=494) · TOTALS 4.8 pts (N=494).

E. RATINGS — AS ISSUED, GROUPED BY RATING DEFINITION

Historical labels are immutable and are not recalculated. Verified pre-kickoff public revisions are identified in the position export with their originals retained. Each band is reported within one engine and one rating definition (the tier_policy_sha256 sealed at issuance); CUT is that band's lower cut point on the selected win probability. N is shown for every band; tiny bands are not averaged away.

ENGINE	DEFINITION	BAND	CUT	N	W-L-P-V	WIN %	MEAN ISSUED P	EMPIRICAL	CAL GAP	BRIER	LOGLOSS	SCORE N
ATS	FOUR-TIER-V1	WEAK	—	13	10-3-0-0	76.9%	0.5080	0.7692	+0.2613	0.24942	0.69199	13
ATS	FOUR-TIER-V1	MODERATE	0.5300	2	1-1-0-0	50.0%	0.5441	0.5000	-0.0441	0.26227	0.71788	2
ATS	FOUR-TIER-V1	STRONG	0.5600	2	1-1-0-0	50.0%	0.5751	0.5000	-0.0751	0.25930	0.71205	2
ATS	FOUR-TIER-V1	ELITE	0.6000	2	2-0-0-0	100.0%	0.8848	1.0000	+0.1152	0.01405	0.12289	2
ATS	SHARED-51.75	WEAK	—	108	58-50-0-0	53.7%	0.5085	0.5370	+0.0285	0.24948	0.69211	108
ATS	SHARED-51.75	MODERATE	0.5175	113	57-56-0-0	50.4%	0.5260	0.5044	-0.0216	0.25122	0.69558	113
ATS	SHARED-51.75	STRONG	0.5375	16	11-5-0-0	68.8%	0.5429	0.6875	+0.1446	0.23663	0.66635	16
ATS	SHARED-51.75	ELITE	0.5575	1	0-1-0-0	0.0%	0.5592	0.0000	-0.5592	0.31272	0.81919	1
ATS	SHARED-52/54/58	WEAK	—	65	33-32-0-0	50.8%	0.5099	0.5077	-0.0023	0.25004	0.69324	65
ATS	SHARED-52/54/58	MODERATE	0.5200	45	18-27-0-0	40.0%	0.5296	0.4000	-0.1296	0.25715	0.70746	45
ATS	SHARED-52/54/58	STRONG	0.5400	15	11-4-0-0	73.3%	0.5486	0.7333	+0.1847	0.22699	0.64693	15
ATS	SHARED-52/54/58	ELITE	0.5800	0	0-0-0-0	N/A	N/A	N/A	N/A	N/A	N/A	0
ATS	ENGINE-V2	WEAK	—	30	13-17-0-0	43.3%	0.5060	0.4333	-0.0727	0.25171	0.69658	30
ATS	ENGINE-V2	MODERATE	0.5120	21	10-11-0-0	47.6%	0.5193	0.4762	-0.0431	0.25157	0.69629	21
ATS	ENGINE-V2	STRONG	0.5318	4	0-4-0-0	0.0%	0.5377	0.0000	-0.5377	0.28908	0.77147	4
ATS	ENGINE-V2	ELITE	0.5460	1	1-0-0-0	100.0%	0.5488	1.0000	+0.4512	0.20355	0.59995	1
ATS	SLATE-QUARTILES-V3	WEAK	—	15	9-6-0-0	60.0%	0.5053	0.6000	+0.0947	0.24674	0.68663	15
ATS	SLATE-QUARTILES-V3	MODERATE	slate-specific	14	10-4-0-0	71.4%	0.5102	0.7143	+0.2041	0.24551	0.68417	14
ATS	SLATE-QUARTILES-V3	STRONG	slate-specific	14	7-7-0-0	50.0%	0.5170	0.5000	-0.0170	0.24958	0.69231	14
ATS	SLATE-QUARTILES-V3	ELITE	slate-specific	13	8-5-0-0	61.5%	0.5261	0.6154	+0.0893	0.24520	0.68355	13
TOTALS	FOUR-TIER-V1	WEAK	—	10	6-4-0-0	60.0%	0.5116	0.6000	+0.0884	0.25087	0.69488	10
TOTALS	FOUR-TIER-V1	MODERATE	0.5300	3	0-3-0-0	0.0%	0.5419	0.0000	-0.5419	0.29367	0.78071	3
TOTALS	FOUR-TIER-V1	STRONG	0.5600	1	0-1-0-0	0.0%	0.5634	0.0000	-0.5634	0.31743	0.82876	1
TOTALS	FOUR-TIER-V1	ELITE	0.6000	5	4-1-0-0	80.0%	0.7078	0.8000	+0.0922	0.21670	0.64984	5
TOTALS	SHARED-51.75	WEAK	—	87	50-37-0-0	57.5%	0.5091	0.5747	+0.0657	0.24878	0.69070	87
TOTALS	SHARED-51.75	MODERATE	0.5175	77	38-39-0-0	49.4%	0.5266	0.4935	-0.0331	0.25186	0.69688	77
TOTALS	SHARED-51.75	STRONG	0.5375	55	26-29-0-0	47.3%	0.5457	0.4727	-0.0730	0.25595	0.70510	55
TOTALS	SHARED-51.75	ELITE	0.5575	19	14-5-0-0	73.7%	0.5761	0.7368	+0.1608	0.22128	0.63525	19
TOTALS	SHARED-52/54/58	WEAK	—	60	32-28-0-0	53.3%	0.5095	0.5333	+0.0238	0.25008	0.69331	60
TOTALS	SHARED-52/54/58	MODERATE	0.5200	40	15-25-0-0	37.5%	0.5277	0.3750	-0.1527	0.25910	0.71137	40
TOTALS	SHARED-52/54/58	STRONG	0.5400	25	11-14-0-0	44.0%	0.5521	0.4400	-0.1121	0.25977	0.71280	25
TOTALS	SHARED-52/54/58	ELITE	0.5800	0	0-0-0-0	N/A	N/A	N/A	N/A	N/A	N/A	0
TOTALS	ENGINE-V2	WEAK	—	23	13-10-0-0	56.5%	0.5028	0.5652	+0.0624	0.24990	0.69295	23
TOTALS	ENGINE-V2	MODERATE	0.5044	22	10-12-0-0	45.5%	0.5071	0.4545	-0.0526	0.25142	0.69599	22
TOTALS	ENGINE-V2	STRONG	0.5111	11	8-3-0-0	72.7%	0.5130	0.7273	+0.2143	0.24356	0.68027	11
TOTALS	ENGINE-V2	ELITE	0.5209	0	0-0-0-0	N/A	N/A	N/A	N/A	N/A	N/A	0
TOTALS	SLATE-QUARTILES-V3	WEAK	—	15	6-9-0-0	40.0%	0.5021	0.4000	-0.1021	0.25023	0.69361	15
TOTALS	SLATE-QUARTILES-V3	MODERATE	slate-specific	14	8-6-0-0	57.1%	0.5042	0.5714	+0.0673	0.24953	0.69220	14
TOTALS	SLATE-QUARTILES-V3	STRONG	slate-specific	14	7-7-0-0	50.0%	0.5082	0.5000	-0.0082	0.24964	0.69242	14
TOTALS	SLATE-QUARTILES-V3	ELITE	slate-specific	13	6-7-0-0	46.2%	0.5122	0.4615	-0.0507	0.25237	0.69789	13

Ratings classify the issued probability; current prospective evidence does not establish monotonic realized performance across rating bands. Spearman rank correlation of band rank with realized W/L, within each engine and definition: ATS FOUR-TIER-V1 -0.053001 (N=19); ATS SHARED-51.75 0.004591 (N=238); ATS SHARED-52/54/58 0.032005 (N=125); ATS ENGINE-V2 -0.041355 (N=56); ATS SLATE-QUARTILES-V3 -0.038553 (N=56); TOTALS FOUR-TIER-V1 0.021071 (N=19); TOTALS SHARED-51.75 -0.014188 (N=238); TOTALS SHARED-52/54/58 -0.109786 (N=125); TOTALS ENGINE-V2 0.061011 (N=56); TOTALS SLATE-QUARTILES-V3 0.031972 (N=56). Descriptive only.

F. INTEGRITY SUMMARY

Counts are computed from the displayed rows. Line, price, probability and market-record checks compare each sealed position with the proposition its immutable grade settled.

CHECK	TARGET SLATE	PHYSICAL CORPUS	EXPECTED	NOTE
ISSUED	2	1020	—	—
SETTLED	2	988	—	—
PENDING	0	0	≥0	official result not final; settles automatically
VOID	0	0	≥0	voided under the applicable FanDuel rule
PUSH	0	0	≥0	line landed exactly on the result
EXCLUDED POST KICKOFF	0	32	≥0	issued after kickoff; visible, never scored
DUPLICATES	0	0	0	—
MISSING GRADES	0	0	≥0	pending rows without a grade object (expected while a game is not final)
MISSING ISSUANCE EVIDENCE	0	0	0	—
LINE MISMATCHES	0	0	0	—
PRICE MISMATCHES	0	0	0	—
PROBABILITY MISMATCHES	0	0	0	—
MARKET RECORD MISMATCHES	0	0	0	—
T2 SNAPSHOT MISMATCHES	0	0	0	grade settled against a T-2 snapshot other than the one sealed on the card
MODEL HASH MISMATCHES	0	0	0	positions whose sealed engine hash differs from their own slate and engine
ENGINE EPOCHS	2	36	—	distinct sealed (engine, hash) pairs; earlier epochs are retained as issued (Appendix 4)
TARGET-SLATE ATS ENGINE = INSTALLED LOCKED ENGINE	MATCH	—	MATCH	—
TARGET-SLATE TOTALS ENGINE = INSTALLED LOCKED ENGINE	MATCH	—	MATCH	—

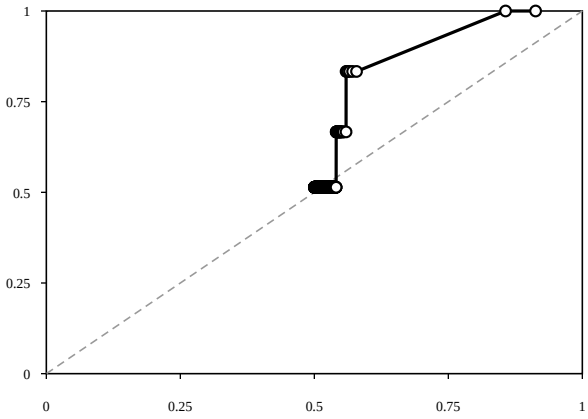
ENGINE_RERUN_COUNT=0 · CLOSING_LINE_SUBSTITUTION_COUNT=0 · MODEL_WRITE_COUNT=0 · ISSUANCE_WRITE_COUNT=0 · SETTLEMENT_WRITE_COUNT=0 · POSTGAME_FEATURE_BACKFILL_COUNT=0

KNOWN EVIDENCE LIMITATIONS=1 · FANDUEL_SETTLEMENT_RULE_SOURCE=UNRECORDED_JURISDICTION_DETAIL (stated under HOW TO READ).

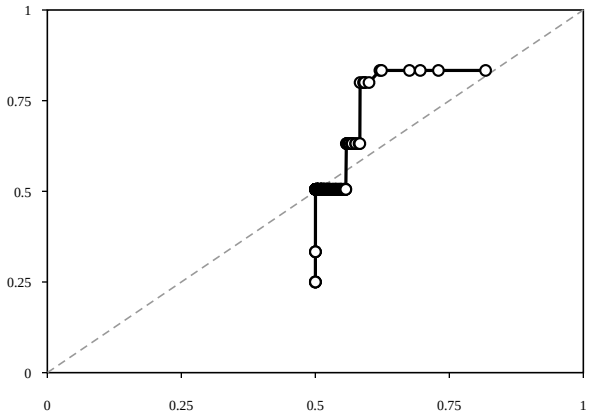
APEX NCAA FOOTBALL - DAILY GRADER — TECHNICAL APPENDIX

APPENDIX 1 — CALIBRATION AND DISCRIMINATION DIAGNOSTICS (RECORD ELIGIBLE W/L ROWS)

ATS CORP / PAV RELIABILITY



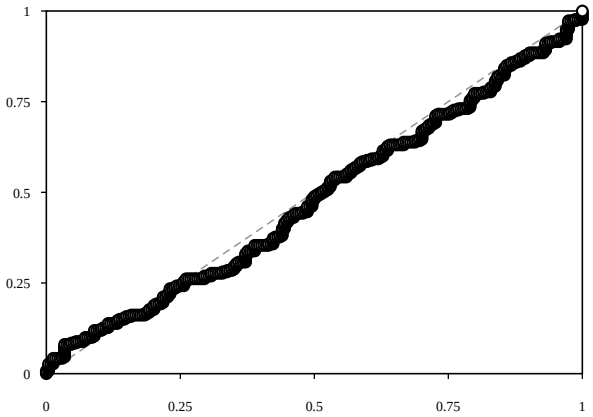
TOTALS CORP / PAV RELIABILITY



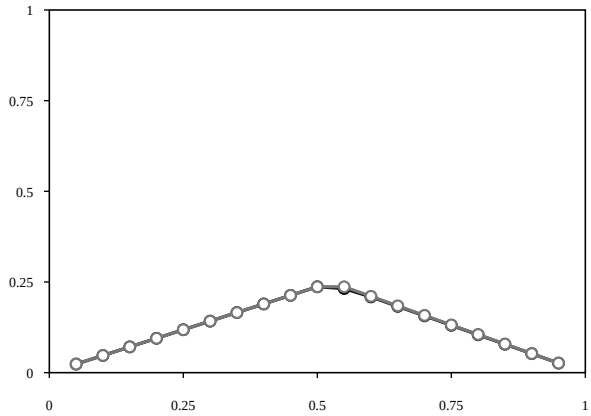
ENGINE	N	BRIER	MISCALIBRATION	DISCRIMINATION	UNCERTAINTY	IDENTITY RESIDUAL
ATS	494	0.2489626	0.0016548	0.0030339	0.2493075	0.0010342
TOTALS	494	0.2509163	0.0019809	0.0034280	0.2497992	0.0025642

CORP/PAV is a reliability diagnostic and proper-score decomposition on N=494 ATS and N=494 TOTALS record eligible W/L rows; descriptive, no significance claim.

ATS ROC



ATS MURPHY



ATS ROC AUC=0.479717 (N=494) · TOTALS AUC=0.466650 (N=494) · discrimination diagnostics only, not proper scores. Murphy: elementary-score / decision-threshold diagnostic, lower is better.

APPENDIX 2 — SLICES: DIRECTION, LINE RANGE, TOTALS DISTRIBUTION

ATS DEFINITION · BAND	FAVORITE N/W/L/P	UNDERDOG N/W/L/P	LOGLOSS	BRIER	SCORE N
FOUR-TIER-V1 · WEAK	8/6/2/0	5/4/1/0	0.69199	0.24942	13
FOUR-TIER-V1 · MODERATE	2/1/1/0	0/0/0/0	0.71788	0.26227	2
FOUR-TIER-V1 · STRONG	0/0/0/0	2/1/1/0	0.71205	0.25930	2
FOUR-TIER-V1 · ELITE	0/0/0/0	2/2/0/0	0.12289	0.01405	2
SHARED-51.75 · WEAK	81/47/34/0	27/11/16/0	0.69211	0.24948	108
SHARED-51.75 · MODERATE	95/49/46/0	18/8/10/0	0.69558	0.25122	113
SHARED-51.75 · STRONG	11/8/3/0	5/3/2/0	0.66635	0.23663	16
SHARED-51.75 · ELITE	0/0/0/0	1/0/1/0	0.81919	0.31272	1
SHARED-52/54/58 · WEAK	41/22/19/0	24/11/13/0	0.69324	0.25004	65
SHARED-52/54/58 · MODERATE	28/10/18/0	17/8/9/0	0.70746	0.25715	45
SHARED-52/54/58 · STRONG	11/8/3/0	4/3/1/0	0.64693	0.22699	15
SHARED-52/54/58 · ELITE	0/0/0/0	0/0/0/0	N/A	N/A	0
ENGINE-V2 · WEAK	19/7/12/0	11/6/5/0	0.69658	0.25171	30
ENGINE-V2 · MODERATE	15/7/8/0	6/3/3/0	0.69629	0.25157	21
ENGINE-V2 · STRONG	4/0/4/0	0/0/0/0	0.77147	0.28908	4
ENGINE-V2 · ELITE	1/1/0/0	0/0/0/0	0.59995	0.20355	1
SLATE-QUARTILES-V3 · WEAK	8/5/3/0	7/4/3/0	0.68663	0.24674	15
SLATE-QUARTILES-V3 · MODERATE	3/2/1/0	11/8/3/0	0.68417	0.24551	14
SLATE-QUARTILES-V3 · STRONG	9/3/6/0	5/4/1/0	0.69231	0.24958	14
SLATE-QUARTILES-V3 · ELITE	8/5/3/0	5/3/2/0	0.68355	0.24520	13

TOTALS DEFINITION · BAND	OVER N/W/L/P	UNDER N/W/L/P	PMF LOGSCORE	RPS	PMF N
FOUR-TIER-V1 · WEAK	2/1/1/0	8/5/3/0	3.907464	0.029001	10
FOUR-TIER-V1 · MODERATE	0/0/0/0	3/0/3/0	4.510078	0.054096	3
FOUR-TIER-V1 · STRONG	0/0/0/0	1/0/1/0	3.951097	0.022986	1
FOUR-TIER-V1 · ELITE	0/0/0/0	5/4/1/0	3.913484	0.024155	5
SHARED-51.75 · WEAK	39/25/14/0	48/25/23/0	4.090637	0.033752	87
SHARED-51.75 · MODERATE	15/12/3/0	62/26/36/0	4.016740	0.032025	77
SHARED-51.75 · STRONG	6/4/2/0	49/22/27/0	4.149179	0.037101	55
SHARED-51.75 · ELITE	0/0/0/0	19/14/5/0	4.052026	0.032149	19
SHARED-52/54/58 · WEAK	39/21/18/0	21/11/10/0	4.088244	0.034117	60
SHARED-52/54/58 · MODERATE	14/5/9/0	26/10/16/0	4.107484	0.036594	40
SHARED-52/54/58 · STRONG	10/6/4/0	15/5/10/0	4.016503	0.031362	25
SHARED-52/54/58 · ELITE	0/0/0/0	0/0/0/0	N/A	N/A	0
ENGINE-V2 · WEAK	16/7/9/0	7/6/1/0	4.074085	0.033889	20
ENGINE-V2 · MODERATE	19/8/11/0	3/2/1/0	4.173835	0.033637	22
ENGINE-V2 · STRONG	11/8/3/0	0/0/0/0	3.860586	0.028300	11
ENGINE-V2 · ELITE	0/0/0/0	0/0/0/0	N/A	N/A	0

TOTALS DEFINITION · BAND	OVER N/W/L/P	UNDER N/W/L/P	PMF LOGSCORE	RPS	PMF N
SLATE-QUARTILES-V3 · WEAK	9/3/6/0	6/3/3/0	4.189288	0.041992	15
SLATE-QUARTILES-V3 · MODERATE	8/6/2/0	6/2/4/0	4.407191	0.044611	14
SLATE-QUARTILES-V3 · STRONG	11/6/5/0	3/1/2/0	4.113432	0.035804	14
SLATE-QUARTILES-V3 · ELITE	13/6/7/0	0/0/0/0	3.816391	0.028162	13

TARGET-SLATE RETAINED TOTALS PMFs=1 (maximum normalization error 1.110e-16) · PROSPECTIVE PMF-SCORED TOTALS POSITIONS=491. No PMF was inferred from moments or binary probabilities.

ENGINE	FAMILY	SLICE	N	W-L-P	WIN %	MEAN P	EMPIRICAL	LOGLOSS	BRIER	SCORE N
ATS	DIRECTION	FAVORITE	344	181-163-0	52.6%	0.5195	0.5262	0.69273	0.24979	344
ATS	DIRECTION	UNDERDOG	150	79-71-0	52.7%	0.5219	0.5267	0.68595	0.24706	150
ATS	SPREAD_SIZE	0-3	65	26-39-0	40.0%	0.5209	0.4000	0.70000	0.25343	65
ATS	SPREAD_SIZE	14+	230	136-94-0	59.1%	0.5218	0.5913	0.68386	0.24579	230
ATS	SPREAD_SIZE	3.5-7	100	47-53-0	47.0%	0.5186	0.4700	0.69821	0.25253	100
ATS	SPREAD_SIZE	7.5-14	99	51-48-0	51.5%	0.5179	0.5152	0.69277	0.24981	99
TOTALS	DIRECTION	OVER	212	118-94-0	55.7%	0.5136	0.5566	0.69021	0.24853	212
TOTALS	DIRECTION	UNDER	282	136-146-0	48.2%	0.5318	0.4823	0.69898	0.25271	282
TOTALS	TOTAL_SIZE	45-52.5	231	131-100-0	56.7%	0.5127	0.5671	0.69110	0.24898	231
TOTALS	TOTAL_SIZE	53-59.5	184	84-100-0	45.7%	0.5297	0.4565	0.69425	0.25068	184
TOTALS	TOTAL_SIZE	60+	41	18-23-0	43.9%	0.5593	0.4390	0.72724	0.26497	41
TOTALS	TOTAL_SIZE	<45	38	21-17-0	55.3%	0.5267	0.5526	0.69040	0.24863	38

APPENDIX 3 — WEEKLY AND ROLLING EVIDENCE

SEASON-WEEK	ENGINE	N	W	L	P	V	PEND	LOGLOSS	BRIER	SCORE N
2026-W1	ATS	98	58	40	0	0	0	0.681548	0.245199	98
2026-W1	TOTALS	98	50	48	0	0	0	0.698386	0.252019	98
2026-W2	ATS	86	47	39	0	0	0	0.691017	0.248935	86
2026-W2	TOTALS	86	46	40	0	0	0	0.690113	0.248491	86
2026-W3	ATS	73	35	38	0	0	0	0.694683	0.250766	73
2026-W3	TOTALS	73	42	31	0	0	0	0.687370	0.247131	73
2026-W4	ATS	54	24	30	0	0	0	0.697022	0.251941	54
2026-W4	TOTALS	54	28	26	0	0	0	0.699773	0.253289	54
2026-W5	ATS	59	32	27	0	0	0	0.689044	0.247959	59
2026-W5	TOTALS	59	24	35	0	0	0	0.705401	0.256117	59
2026-W6	ATS	58	25	33	0	0	0	0.698314	0.252583	58
2026-W6	TOTALS	58	30	28	0	0	0	0.695412	0.251131	58
2026-W7	ATS	62	35	27	0	0	0	0.690855	0.248854	62
2026-W7	TOTALS	62	31	31	0	0	0	0.693489	0.250171	62
2026-W8	ATS	4	4	0	0	0	0	0.658397	0.232633	4
2026-W8	TOTALS	4	3	1	0	0	0	0.682752	0.244803	4

WINDOW	STATUS	SLATES	N	W	L	P	V	PENDING
ROLLING_LATEST_5	AVAILABLE	5	124	67	57	0	0	0
ROLLING_LATEST_10	AVAILABLE	10	242	124	118	0	0	0
ROLLING_LATEST_20	AVAILABLE	20	618	311	307	0	0	0

APPENDIX 4 — CORPUS COVERAGE, EPOCHS AND HASHES

SLATE	PHYSICAL	EXCLUDED	DIRECTIONAL	EXACT P	EXACT PMF	FEATURE VECTOR	PENDING	VOID
2026-08-29	16	0	16	16	16	0	0	0
2026-09-03	22	0	22	22	22	22	0	0
2026-09-04	14	0	14	14	14	14	0	0
2026-09-05	136	0	136	136	136	136	0	0
2026-09-06	6	0	6	6	6	6	0	0
2026-09-07	2	0	2	2	2	2	0	0
2026-09-10	2	0	2	2	2	2	0	0
2026-09-11	10	0	10	10	10	10	0	0
2026-09-12	160	0	160	160	160	160	0	0
2026-09-17	2	0	2	2	2	2	0	0
2026-09-18	6	0	6	6	6	6	0	0
2026-09-19	138	0	138	138	138	138	0	0
2026-09-24	2	0	2	2	2	2	0	0
2026-09-25	10	2	10	10	10	10	0	0
2026-09-26	128	30	128	128	128	128	0	0
2026-10-01	4	0	4	4	4	4	0	0
2026-10-02	6	0	6	6	6	6	0	0
2026-10-03	108	0	108	108	108	108	0	0
2026-10-06	2	0	2	2	2	2	0	0
2026-10-07	4	0	4	4	4	4	0	0
2026-10-08	8	0	8	8	8	8	0	0
2026-10-09	10	0	10	10	10	10	0	0
2026-10-10	92	0	92	92	92	92	0	0
2026-10-13	4	0	4	4	4	4	0	0
2026-10-14	4	0	4	4	4	4	0	0
2026-10-15	6	0	6	6	0	0	0	0
2026-10-16	6	0	6	6	6	6	0	0
2026-10-17	104	0	104	104	104	104	0	0
2026-10-20	6	0	6	6	6	6	0	0
2026-10-21	2	0	2	2	2	2	0	0

Only physical immutable production evidence is listed; no slate is synthesized. EXCLUDED counts positions issued post-kickoff on that slate.

SLATE	ATS MODEL	ATS SHA256	TOTALS MODEL	TOTALS SHA256	ISSUANCE SHA256	POS
2026-08-29	GLOBAL_EMPIRICAL_V1	2ac5790dca0f91da5431f44fb4262ef0398f2adb1897cb9ded311fb4470c8798	RIDGE_GAUSSIAN_A100_V1	7524bfe56bc65257e5257a5e751540ed29bb6ca027b5193af87db35aaa79da9b	7ca849afef6c528d1ead8d7437d69de4c2d0495fa17ae6a7cc149b752b7fb5cf	16
2026-09-03	D2	fc79ff17b8de2a00019e10f4dc8c2f5bb16acf14648fcb65a31decc3d9368494	TOTALS-V5-RECONCILED-POOL	d22ea083819b70c22f409a032ddab1d03322c10790541c8d458266868e7835e8	206e29bdc0684f4fb28c38e4d31fef213044b0e2b5921244ee6c95f59bf577f5	22
2026-09-04	D2	61bde29368eeadbad93677743726c0ac7e6ae58e1607ca0efeae6be28791af8a	TOTALS-V5-RECONCILED-POOL	1949c669883a6de379e866a7955ba414526a74edcecd9980f56cb67774ba15a4	60d17f19f198dacf58b27b9effa827a2a218da8ee50d8167358bc6e5a96ff77b	14
2026-09-05	D2	fbaf65a5d6c42a1dd3eb896fa6448101ac18ae87a2c725208597639a13f8c28a	TOTALS-V5-RECONCILED-POOL	a44b3556bb82badf472033ad582e3872b74eef8c16b880dd83bc564cb866393e	d4e843d2045430ab571451618707fc263e098730c11387f78f18216d8a9db317	136
2026-09-06	D2	6b1b00207e63d1bfc02fb5b0e0a8e10d43d65b75e507b5d779281598306d03c4	TOTALS-V5-RECONCILED-POOL	3c8333e071f531230eb6df70c771efca66e2502be153f12860590952c88e104f	86db9039d44d631ce1b1c4d2634b245ee4de57f6e7c16320bb0b4c1b0eb57638	6
2026-09-07	D2	9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7	TOTALS-V5-RECONCILED-POOL	4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3	21ae7f8b7c21dc9dd8a2648037b759e41932cd41a56ace606a6c750071a6ba2f	2
2026-09-10	D2	9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7	TOTALS-V5-RECONCILED-POOL	4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3	092c007808e9eaf0621ad447487a4eb2edcc6b357dded58966dd80a699905465	2
2026-09-11	D2	9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7	TOTALS-V5-RECONCILED-POOL	4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3	31dbf89d125019d978fc51ede5e2901cd580547ed9a92cb061dae348a880ebd9	10
2026-09-12	D2	9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7	TOTALS-V5-RECONCILED-POOL	4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3	ef2c885600d42f8ab9638655b55f6e660784633a619de2f4a2689f704cf1833a	160
2026-09-17	D2	9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7	TOTALS-V5-RECONCILED-POOL	4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3	cd22addc00b553ae62058953be1695202952efdf5db43f4a47a6bd1993fbf491	2
2026-09-18	D2	9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7	TOTALS-V5-RECONCILED-POOL	4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3	a3ca7166267b456484096417ec3d0d10f67ec1fac36800fb4f3eee6280b70981	6
2026-09-19	D2	9d2e7e3fd1d851c4acfd6cae6c5936a6e2640f2ec47c1b9a6fd887ddda2505e7	TOTALS-V5-RECONCILED-POOL	4867d2e67a77260058b42c84ce61f53000ecebc861a72c0d226cc52261dd75c3	0b08644781 2f9a3953a6dd7df666703859be26ec5e4c13f3d54469bc31c53e69	138
2026-09-24	D2	e46880f3483a7badb7329fee2c709eb1b63b1a250aea9dfaa2cd2fdc63606469	TOTALS-V5-RECONCILED-POOL	4a9b487849a1899e8ad11ff10fbce1e8c16243e6d8accdb95ff4de5d7cf26725	aa1effee193bc769a1faa9561f968673c5f34c0767d22184b3b6cafdceec0f9c	2
2026-09-25	D2	e46880f3483a7badb7329fee2c709eb1b63b1a250aea9dfaa2cd2fdc63606469	TOTALS-V5-RECONCILED-POOL	4a9b487849a1899e8ad11ff10fbce1e8c16243e6d8accdb95ff4de5d7cf26725	f54aa7358ceb2c8c587ae954f77dd2316bf608a094e612857102cc0fa6c8bb50	10
2026-09-26	D2	69d8a6c392ded2d2fb728b95ab69255b505709bce7cbfd9387314cefe666359c	TOTALS-V5-RECONCILED-POOL	43a2ac80cd611e14793cdbd76d3c663457ea4f78cdfe9d4aed4ce0be30292f4b	bdcffed33a92cbe8a811365359aa59e800fb5764b4d85ca4a734bb40c46dc9c8	128
2026-10-01	D2	6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307501f9969	TOTALS-V5-RECONCILED-POOL	a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3	dbeb13faabe6375ba5f07cf107e1f8d498c2da288a0fab60cf9f09c029573f0c	4
2026-10-02	D2	6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307501f9969	TOTALS-V5-RECONCILED-POOL	a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3	67b8ec6ef73f4a059133509d82312dd2d03be72013f901cedff7a266e5b9b5d2	6
2026-10-03	D2	6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307501f9969	TOTALS-V5-RECONCILED-POOL	a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3	96ca5988359bc3786e4d2f39ec67da17fcf15cd3a11acca9da27f19942248637	108
2026-10-06	D2	6553b110ab1f5610bb3e661030b51219729111e8bb17b0ec35b306307501f9969	TOTALS-V5-RECONCILED-POOL	a96560915ec8fe89648a872c06b463bf1e8ff911181210ca27661083e83e53f3	a6f873de88fad0362a04c348c70cc6db1212c361f0db7e4ebb7f3460964bdaf7	2
2026-10-07	D2	83c82036b5009db103d32cfe38078c5f906d60f076155bd7493f6b722c151c3d	TOTALS-V5-RECONCILED-POOL	699cbaa1ab0a58c1ee0c215beaf1b59d093ff934c15733141b76ac725f890a7c	6e6c1a1868dde2fdaebcc7edd866c3eb2893f0e79eabfbd1f3cca6210e18c0a3	4
2026-10-08	D2	0e60ebc6cde767d75ce8811b2b3a660229145af0c76868a62f78b685ea654a33	TOTALS-V5-RECONCILED-POOL	9f586104744d101077a66b29a2bc3d8cec8ac28aa9bd5d5e003fb577b5bd71d2	5d4a2697f56d9d6387da5120ad482db3cdea2fa0e201c0d8dab08d491973c4b6	8
2026-10-09	D2	39e779ec7f112d488a9379cac49ae70553895dfa2027df849a91ef512fcde8aa	TOTALS-V5-RECONCILED-POOL	6a90e31cb2e12b0af43b74ce3a1f7cc52cf61599394cbb3f90d89a753f50de81	bec1312906c155872704c3df9cf35c1ac0940cd28254caf7177a0b533873885b	10
2026-10-10	D2	3844cb449793385ad6c30e02be76e8247ab5eb8f421fd8ac4631c4bdabc1d45c	TOTALS-V5-RECONCILED-POOL	8e3f0746c12e8459e8a70ea4e41a36dc2fbccc085d7f7c93f38ce14eda497d26	7f0a3507742b30cf9f187b3f1a482158124a8bba747eaf80d8cc4e86c2745979	92
2026-10-13	D2	3844cb449793385ad6c30e02be76e8247ab5eb8f421fd8ac4631c4bdabc1d45c	TOTALS-V5-RECONCILED-POOL	8e3f0746c12e8459e8a70ea4e41a36dc2fbccc085d7f7c93f38ce14eda497d26	e0ff93a241d1e5a9926e60356a0baecc8983d89728cccc4e2db5c8f7144555e2	4
2026-10-14	D2	3844cb449793385ad6c30e02be76e8247ab5eb8f421fd8ac4631c4bdabc1d45c	TOTALS-V5-RECONCILED-POOL	8e3f0746c12e8459e8a70ea4e41a36dc2fbccc085d7f7c93f38ce14eda497d26	1152153c9bdbbb876eff7e6fe0748570a39b49f50a17b63a45b286befe26ba89	4
2026-10-15	D2	—	TOTALS-V5-RECONCILED-POOL	—	4ca57a804c77b104cf449e21460e68aea8faadead00b6bfa7b7fb5428ffdd11d	6
2026-10-16	D2	c3c63d810d7b3d4b909db08afcb0a32cebdabb523f9488d5b61fb8f6854d9941	TOTALS-V5-RECONCILED-POOL	09bb7f9a3e2c3df5fae7d8d4893038f743a731eaefd5e3db6a595c391541e72f	d36c170767b2eaf6a5171d1ffd6c524d145b02d1ca7809473bfb58cea495c4ae	6
2026-10-17	D2	3bcb42aa283aff56f3a7b81743bffcbeeb16201ef4fee6577b3528cdb0761698	TOTALS-V5-RECONCILED-POOL	ad4a7acc8fc64de5411af3cc3ff8395ad2d978aaee857ca5599323c276e8d029	4e324d5c8eb5773e15ddf6ae0a38193a2ed42471a588b49713d2e0541d403d07	104
2026-10-20	D2	09c33b70af53b0a53e2127b16b00d23d37b8de692edc76f1ab83b1a25109d2e0	TOTALS-V5-RECONCILED-POOL	e66c7f0d35a0827a93458b84e083d3d0fa4e54724af82470a674acb961c2243f	0d15a3c6f93c346e56541272336810335 08f8c3a2362d4e2576a3a16b873348c	6
2026-10-21	D2	d50e21788ae9f7107b8641e1e8e89daad26b3e828b85994137d2fe576346fd19	TOTALS-V5-RECONCILED-POOL	3a38517a5262c5e818e8f3222ff0811d748e4c6e03cccc7950ffecc626149980	3840287e23fc5e6008b90b4d6b390ffe391c9b85b60ce8f23759df3897315107	2

Each slate reports the engines physically sealed on that slate; earlier epochs are not relabelled with later champion identities.

APPENDIX 5 — ISSUE-TIME EVIDENCE COVERAGE, CADENCE, DATA / CONTRACT RECEIPT AND SEQUENTIAL EVIDENCE

OBJECT (TARGET SLATE)	CAPTURED	PHYSICAL COUNT	LIMITATION
ISSUED ATS PROBABILITY	YES	1	NONE
TOTALS FULL PMF	YES	1	RETAINED IN SEALED CARD
MARKET LINE / ISSUED PRICE	YES	2	NONE
FANDUEL QUOTE AT ISSUANCE (record sha, timestamps)	YES	2	NONE
MODEL ID / HASH · AS-ISSUED RATING	YES	2	NONE
D2 / V5 EXACT VECTOR	YES	1	NONE
T3 / T2 SNAPSHOT HASHES	YES	1	NONE
ISSUE-TIME FEATURE SIDECAR	YES	1	NONE
SETTLEMENT	YES	2	IMMUTABLE GRADE AUTHORITY; POST-KICKOFF EXCLUDED

D2_INPUTS_FILLED=270/270 · V5_INPUTS_FILLED=223/223 (filled/registered inputs the engines read, summed over this slate's champion cards). Champion cards retain exact vectors, missingness, source timestamps, availability state, full PMFs and FanDuel comparators; FCS fields the provider does not publish are retained as explicit missingness.

CADENCE	PERMANENT NCAA REVIEW ROLE
DAILY	Immutable issuance-to-grade reconciliation; report products; delivery ledger
WEEKLY	Records, proper scores, direction and line-range slices with N
MONTHLY	Calibration, discrimination, band order, hash epochs and coverage
QUARTERLY	Scientific defensibility and prospective evidence review; no automatic retraining

OBJECT	PATH / ID · ROWS · SCHEMA	SHA256
SEALED ISSUANCE	/var/opt/apex_ncaaf/production/issuance/2026-10-21/slate.json	3840287e23fc5e6008b90b4d6b390ffe391c9b85b60ce8f23759df3897315107
GRADE COMPLETION	/var/opt/apex_ncaaf/production/grader_completion_state/2026-10-21.json	7a5eca5899a06278789c8bd281a272d790fe38696b92e0590f42a317f96e4a17
ATS MODEL AT ISSUE	D2	d50e21788ae9f7107b8641e1e8e89daad26b3e828b85994137d2fe576346fd19
TOTALS MODEL AT ISSUE	TOTALS-V5-RECONCILED-POOL	3a38517a5262c5e818e8f3222ff0811d748e4c6e03cccc7950ffecc626149980
POSITION EVIDENCE	/var/opt/apex_ncaaf/production/results/2026-10-21/evidence/ncaaf_position_evidence.csv	7da121647a5b1c9615efb73cbe273b7c20fe5f2470bf5a97ed4c7231b7b9afb5
COMPLETE SCIENTIFIC SIDECAR	/var/opt/apex_ncaaf/production/results/2026-10-21/evidence/ncaaf_complete_scientific_evidence.json · 1014 · apex.ncaaf.complete_position_scientific_evidence.2026-08-31.v1	7c4e7afe5a2c682b654298c07f11bbabee3340849e3b4e4211987c8b36816ecf
SEQUENTIAL FANDUEL SIDECAR	/var/opt/apex_ncaaf/production/results/2026-10-21/evidence/ncaaf_prospective_fanduel_sequential_evidence.json · 1904 · apex.ncaaf.prospective_fanduel_anytime_evidence.2026-09-04.v2	76e00de99c5abf36f0f19580075f2c7c97423e37f0389124cb45256aca57bd1b
COVERAGE LEDGER	/var/opt/apex_ncaaf/production/results/2026-10-21/evidence/ncaaf_coverage_ledger.csv	19baf47950e55db7a061e5adec644772278da68fe850e2ed80571016c9eec2b5

Brier (1950), DOI 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2

Murphy (1969), DOI 10.1175/1520-0450(1969)008<0988:OTPS>2.0.CO;2

Gneiting & Raftery (2007), DOI 10.1198/016214506000001437

Dimitriadis, Gneiting & Jordan (2021), DOI 10.1073/PNAS.2016191118

Dimitriadis, Gneiting, Jordan & Vogel, DOI 10.1016/j.ijforecast.2023.09.007

Fawcett (2006), DOI 10.1016/j.patrec.2005.10.010

PROSPECTIVE SEQUENTIAL FANDUEL EVIDENCE — CONFIDENCE SEQUENCE / E-PROCESS (RETAINED; INTERVALS NOT YET PRACTICALLY INFORMATIVE)

ENGINE	METRIC	N	MEAN PAIRED ADVANTAGE (FD loss − APEX loss)	95% CS LOW	95% CS HIGH	ANYTIME P
D2	BRIER	476	+0.000638	-0.258820	+0.260097	1.0000
D2	LOGLOSS	476	+0.001279	-8.960089	+8.962647	1.0000
TOTALS-V5-RECONCILED-POOL	BRIER	476	-0.002098	-0.261556	+0.257360	1.0000
TOTALS-V5-RECONCILED-POOL	LOGLOSS	476	-0.004195	-8.965563	+8.957173	1.0000

Never-reset bounded confidence sequence and mixture e-process over every grade the canonical prospective court retains: 952 sealed predictions, 952 graded, 1904 sequence rows, activation epoch 2026-09-01T03:30:00Z. The court admits only positions sealed on or after its activation epoch, so its N per engine is smaller than the paired N in section C, which pairs every record-eligible grade that carries a FanDuel comparator. Framework: BOUNDED_HOEFFDING_STITCHED_CONFIDENCE_SEQUENCE_AND_PREDICTABLE_MIXTURE_E_PROCESS.

APPENDIX 6 — RECORD BY EDGE GROUP AND CLOSING LINE [R4][R5][R10] [R8][R9] standard practice

ENGINE	EDGE GROUP	N	W-L	WIN %	UNITS	RETURN	EDGE RANGE
ATS	TOP 10%	48	30-18	62.5%	+9.13	19.0%	+0.0292 … +0.0572
ATS	10-25%	73	39-34	53.4%	+1.17	1.6%	+0.0215 … +0.0291
ATS	25-50%	122	60-62	49.2%	-8.21	-6.7%	+0.0120 … +0.0214
ATS	BOTTOM 50%	243	126-117	51.9%	-6.17	-2.5%	-0.0210 … +0.0120
TOTALS	TOP 10%	48	23-25	47.9%	-3.73	-7.8%	+0.0512 … +0.1321
TOTALS	10-25%	73	32-41	43.8%	-11.93	-16.3%	+0.0341 … +0.0509
TOTALS	25-50%	122	57-65	46.7%	-12.73	-10.4%	+0.0177 … +0.0336
TOTALS	BOTTOM 50%	243	136-107	56.0%	+14.63	6.0%	-0.0150 … +0.0175

MKT	SELECTION	ISSUED	CLOSE	PTS GAINED	CLOSE FD P	RES
ATS	New Mexico State	-2.5	-3.5	+1.0	0.4850	W

MKT	SELECTION	ISSUED	CLOSE	PTS GAINED	CLOSE FD P	RES
TOTALS	OVER	46.5	46.5	+0.0	0.5150	W

APPENDIX 7 — SOURCES FOR THE RESEARCH LINES

Tags in brackets above point here. References are taken only from the SciSpace reviews supplied to APEX (see RESEARCH_SOURCES.md). 'Standard practice' means no study in those reviews measured it directly.

[R1] Converting College Football Point Spread Differentials to Probabilities. arXiv, Dec 2022. doi:10.48550/arxiv.2212.08116 (cited for the 52.38% break-even at -110 in SciSpace review 'Minimum Edge Requirements...', section 3.3.1).

[R2] A. Fodor, M. DiFilippo, K. Krieger, J. L. Davis (2013). Inefficient pricing from holdover bias in NFL point spread markets. Applied Financial Economics 23(17):1407-1418. doi:10.1080/09603107.2013.829201.

[R3] E. H. Moore, J. Francisco (2020). The SEC vs. the Dow Jones: A Profitable Betting Strategy in NCAA Football. The Journal of Prediction Markets 13(2):3-14. doi:10.5750/JPM.V13I2.1740.

[R4] M. P. Mirabile, M. D. Witte (2016). College football and the Vegas line: Deconstruction and arbitrage. The Journal of Prediction Markets 10(1):53-67. doi:10.5750/JPM.V10I1.1183.

[R5] SciSpace systematic review (2026). Minimum Edge Requirements, Probability Calibration, and Skill Detection for Quantitative NCAA Football Betting Models (24 studies). Selective betting 55-59%; ~1,200 bets to detect a 55% bettor (section 3.3.3, power analysis, no single study cited).

[R6] M. P. Naeini, G. F. Cooper, M. Hauskrecht (2014). Binary Classifier Calibration: Non-parametric approach. arXiv: Machine Learning. (Expected Calibration Error.)

[R7] C. Walsh, A. Joshi (2024). Machine learning for sports betting: Should model selection be based on accuracy or calibration? Machine Learning with Applications, 100539. doi:10.1016/j.mlwa.2024.100539.

[R8] S. Mani (2018). NFL Betting Market Efficiency: A Closer Look at the Final Day of Betting. (Compares opening and closing lines; the SciSpace evaluation-metrics review, 10 studies, found no study that formally measured closing-line value, so CLV here is standard practice.)

[R9] SciSpace systematic review (2026). Evaluation Metrics for American Football Point Spread and Game Totals Forecasting Models (10 studies). Recommended order: closing-line value, return, win rate vs 52.38%, calibration, log loss; at least 500 bets.

[R10] SciSpace systematic review (2026). Predicting NCAA College Football Spreads and Totals (6 studies; no full texts). Selective prediction on the top 8-35% of disagreements.

[R11] SciSpace systematic review (2026). Market Efficiency in NFL and NCAA Football Betting Markets (42 studies).

[R12] R. Borghesi (2008). Applied Financial Economics 18(12):947-953. doi:10.1080/09603100701335432 (weather and totals; cited in shadow/RESULTS.md; weather shadow not run daily because it showed no gain in 2026).

BLACK_AND_WHITE_ONLY=YES · MODEL_WRITE_COUNT=0 · ISSUANCE_WRITE_COUNT=0 · SETTLEMENT_WRITE_COUNT=0 · ENGINE_RERUN_COUNT=0 · CLOSING_LINE_SUBSTITUTION_COUNT=0